Low-Voltage Slippery Microbubble Manipulation Using Dual High-*k* and Lubricating Layers for Ultrasound Modulation

[1]Subeen Kim, [1]Sangho Bang, [2]Gunoh Lee, [2]Kyung Jin Lee, [1]Hyunjoo Jenny Lee*
[1]School of Electrical Engineering, Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea, [2]Department of Chemical Engineering and Applied Chemistry, Chungnam National University (CNU), Daejeon, Republic of Korea
E-mail: tn48619@kaist.ac.kr

이중 고유전율·윤활층을 이용한 저전압 미끄럼 기반 미세기포 제어 및 초음파 변조

[1] 김수빈, [1] 방상호, [2] 이건오, [2] 이경진, [1] 이현주*
[1] 한국과학기술원, [2] 충남대학교
E-mail: tn48619@kaist.ac.kr

Abstract

Real-time acoustic beamforming is essential for remote particle manipulation, cell assembly, and neuromodulation. However, existing phased array systems face challenges with scalable pixel-level addressing, as fully addressing N^2 pixels requires a vast number of interconnections with N^2 signal lines, which limits miniaturization. We present an electrowetting-on-dielectric (EWOD) spatial ultrasound modulator (SUM) that converts plane waves into acoustic images by binary amplitude modulation using microbubble manipulation. A high-*k* dielectric stack (Parylene-OH and Al_2O_3) ensures stable operation in the wet environment with a voltage of 5.2 V_{rms}. A silicone-oil-infused nanoparticle coating prevents bubble pinning, which demonstrates continuous and reversible motion. Using a row-column addressed array, we achieve real-time 2D beamforming of N^2 pixels with only 2N signal lines. This platform paves the way for scalable ultrasound applications in neuromodulation, acoustic tweezers, and cell assembly.

Keywords: ***Electrowetting-on-dielectric, bubble generation, bubble manipulation, acoustic beamforming, spatial ultrasound modulation***

1. Introduction

Precise acoustic beamforming is crucial for remote particle manipulation, directed assembly, fluidic control, and neuromodulation [1]. Traditional methods, such as static acoustic lenses, offer high-resolution beamforming without electronic interfaces, while dynamic phased arrays provide low-latency beam steering but require complex and bulky circuitry. Recent wearable CMOS-based phased arrays face power efficiency and resolution trade-offs [2]. To overcome these issues, dynamic SUMs using microbubble patterns generated by CMOS or DMD devices have been explored. However, these systems are still bulky, which limits their application in wearables [3]. Therefore, a miniaturized dynamic SUM with high spatiotemporal resolution is required.

This work presents a dynamic SUM based on EWOD, which precisely modulates microbubbles with micron-scale electrodes in a liquid chamber under low operation voltage and high reusability. The system is compact (<1 mm thick) and compatible with single-element ultrasound transducers, expanding the utilization in various applications.

2. Design and fabrication

2.1 System description

The miniaturized EWOD spatial ultrasound modulator (SUM) integrates with a single-element transducer and consists of top and bottom substrates separated by a thin liquid gap (Fig. 1). Pixelated bottom electrodes provide EWOD actuation, while aligned top electrodes perform as a common ground and local microbubble generators. Microbubbles are produced by electrolysis on the top, enlarged to contact the hydrophobic bottom surface, and then transported by EWOD-induced contact-angle changes for pixel-level positioning. Because of the large air–water acoustic impedance mismatch (×3,500), bubble-covered pixels block ultrasound while bubble-free pixels transmit the acoustic beam (Fig. 2). Therefore, programmable bubble patterns reshape an incident plane wave into the desired acoustic wavefront at the target plane.

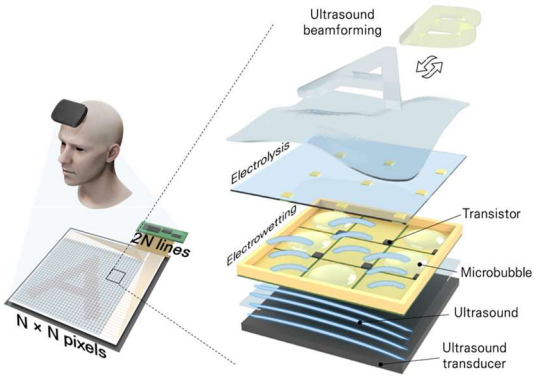

Fig. 1. Schematic illustration of the ultrasound transducer and EWOD-based SUM system.

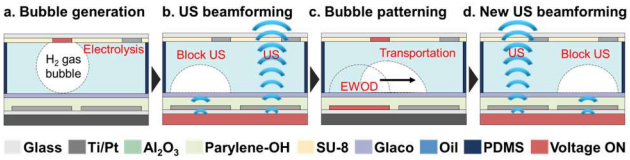

Fig. 2. Schematic diagram of the actuation mechanism.

2.2 Fabrication

The EWOD-based SUM was fabricated to ensure robust insulation and low pinning in the wet operation. For the bottom substrate, Ti (10 nm) and Pt (100 nm) were deposited on the glass substrate and patterned via lift-off. A 50-nm Al_2O_3 high-*k* dielectric layer was sputtered and annealed at 400°C, followed by deposition of 1-µm Parylene-OH by CVD. A 50-nm nanoparticle layer was spray-coated and then infused with silicone oil by dip coating to enhance lubrication. For the top substrate, Ti (10 nm) and Pt (100 nm) were deposited and patterned similarly, and regions where the metal should remain unexposed were passivated with SU-8 using photolithography. Both substrates were thinned by CMP, and contact pads were connected to a PCB. Finally, a PDMS wall (500 nm) was placed on the bottom substrate and filled with 100 mM KCl solution, which was covered by the top substrate.

3. Measurements and results

3.1 Device characterization

For an effective hydrophobic layer in the system, we evaluated four materials: Parylene-F, CYTOP, silica nanoparticle, and silicone-oil–infused silica nanoparticle. Advancing/receding angle measurements on a tilted stage and voltage application revealed that the silicone-oil-infused nanoparticle surface exhibited much smaller contact-angle hysteresis (~3°) than other layers. The hydrophobic layer indicated strong suppression of contact-line pinning and enhanced repeatability of bubble manipulation following voltage application and removal.

Furthermore, the higher permittivity of an Al_2O_3/Parylene-OH double insulator enables the same contact-angle change at lower voltage compared to a single Parylene-C layer. We adopted a 740-nm Parylene-OH layer with the silicone oil-infused nanoparticle coating to balance low-voltage actuation and wet-environment reliability, which achieved a 10° contact-angle change at 21 V_{rms} (Fig. 3).

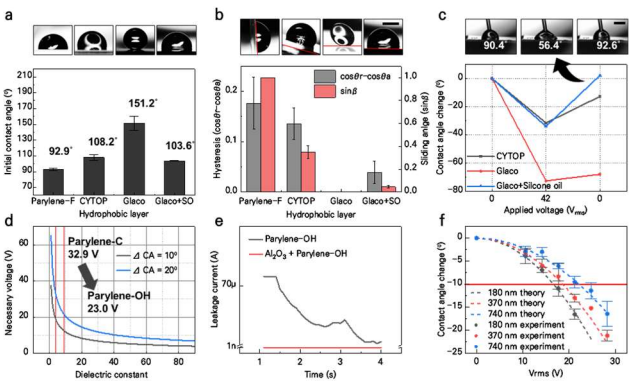

Fig. 3. Electrowetting characteristics of the fabricated substrate.

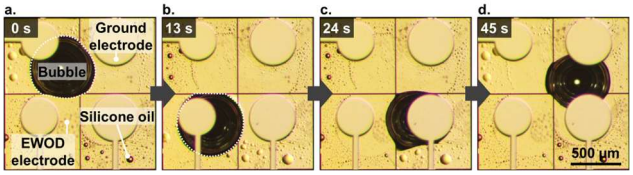

Fig. 4. Microbubble manipulation using EWOD.

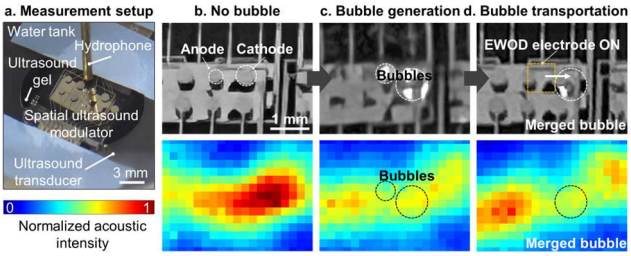

Fig. 5. Optical microscope images of microbubble manipulation and spatial ultrasound modulation.

3.2 Microbubble and ultrasound modulation

Microbubbles generated on the top substrate and expanded to contact the bottom surface upon 3 V. The microbubble manipulation by EWOD was well demonstrated at 5.2 V_{rms} (Fig. 4). Ultrasound measurements confirmed that relocating bubble patterns via EWOD produced distinct sound-pressure profiles, which implies programmable, pattern-dependent modulation of the transmitted acoustic field (Fig. 5).

4. Conclusion

This work demonstrates a compact EWOD-driven dynamic ultrasound modulator by electrically positioning microbubbles, which offers a practical alternative to wiring-intensive phased arrays. An Al_2O_3/Parylene-OH dielectric stack enables reliable low-voltage operation in liquid, while a silicone-oil–infused nanoparticle coating supports repeatable, pinning-free microbubble transportation. With row–column addressing, the device achieves low-power beam control with the single-element transducer. These advancements highlight the versatility and applicability of the EWOD-based spatial ultrasound modulator in next-generation biomedical technologies such as ultrasound neuromodulation.

Acknowledgments

This work was supported by the K-Brain Project of the National Research Foundation (NRF) funded by the Korean government (MSIT) (RS-2023-00262568), Republic of Korea.

References

1. K. Melde et al., Ultrasound-Assisted Tissue Engineering, *Nat. Rev. Bioeng.* **2**, 486-500 (2024).
2. T. Costa, C. Shi, K. Tien, J. M. Elloian, F. A. Cardoso, K. L. Shepard, An Integrated 2D Ultrasound Phased Arr ay Transmitter in CMOS with Pixel Pitch-Matched Beam forming, *IEEE Trans. Biomed. Circuits Syst.* **15**, 731-742 (2021).
3. Z. Ma et al., Spatial Ultrasound Modulation by Digitally Controlling Microbubble Arrays, *Nat. Commun.* **11**, 4537 (2020).